Swami Keshvanand Institute of Technology, Management & Gramothan,
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3CS4-05 - Data Structure and Algorithm

Introduction - Data Structure, Algorithm, Basic Terminology, Data Structure operations, Algorithm Complexity, Time - space Tradeoff

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Data

Data are simply values or sets of values A data item refers to a single unit of values. Data item that are divided into subitems are called group item, else elementary items. An entity is something that has certain attributes The term "Information" means meaningful or process data.

Data - Ram, $, 99,

Name : first name : middle name : Last

Group item

age :

elementry item
or
entity

Information - Ram has $99.

Record - data has organized into the hierarchy of field

file - collection of record

The study of Data structure.

1. Logical & mathematical description of the structure.
2. Implementation of the structure on a Computer
3. Quantitative analysis of structure, (Amt. of mry)

Data Structure

Defn - Data may be organized in many different ways, the logical or mathematical model of a particular organization of data is called a data structure.

or in a simply way - How we are storing data in memory.

Data structures deals with the study of methods, technique, and tools to organize or structure data.

Algorithm -

An algorithm may be defined as a finite sequence of instruction each of which has a clear meaning and can

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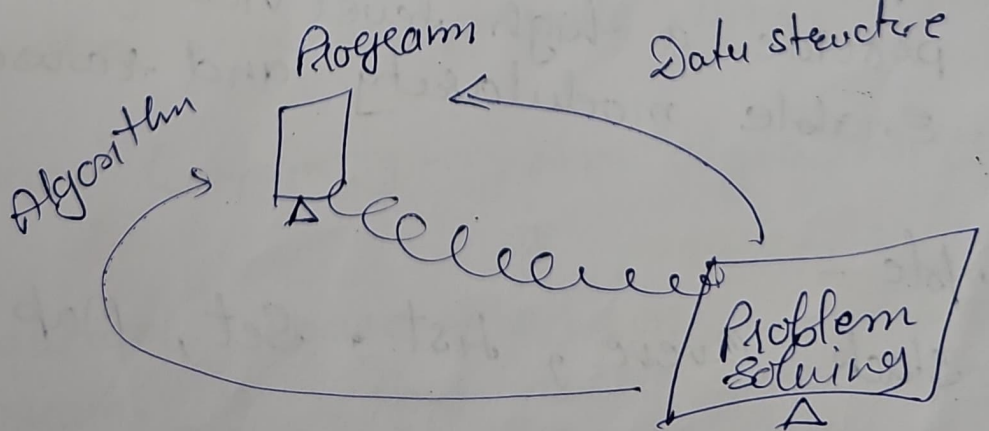
can be performed with a finite amount of effort in a finite length of time.

An algorithm has the following structure

(1) Input step
(2) Decision
(3) Repetitive step
4) Output step

Property -

- finiteness
- definiteness
- effectiveness
- Generality
- Input-output



Abstract data type (ADT)

An abstract data type is a high level description of a data structure that define its behavior and operations without worrying about its implementation specifying details.

* Interface *

ADT {
    Data elements
    operation ()
    Behaviour ()
}

ADTs are useful because

1. separate interface from implementation
2. provide a high level view
3. Enable modularity and reuse

Example -
stack, queue, list, Set, Map

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Analysis of Algorithm

The time and space are two major measure of the efficiency of an algorithm.

Complexity - The Complexity of an algorithm is the function which gives the running time and/or space in terms of the input size.

Time Complexity - An fastest algorithm for the problem or an Algorithm which perform its task in the minimum possible time.

Space Complexity - An algorithm or a program's required space to run a complete program. or How much space required by the program for complete execution.

Prog 1

```
. . . .
. . . .
x = x+2          1
. . .
. . .
```

time = 'c'

Total frequency count = 1

$O(1)$

Prog 2

```
. . . . .
. . . . .
for k = 1 to n do        n+1
    x = x+2              n
end
. . . .
. . . .
```

time = 2n+1

Total freq. count = 2n+1

$O(n)$

Prog 3

```
. . . . .
. . . . .
for k = 1 to n do            n+1
    for j = 1 to n do        n.n+1
        x = x+2              n^2
    end
end
```

Total freq count = $2n^2+n+1$

$O(n^2)$

Big O O - Order of magnitude

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Asymptotic Notations

Asymptotic Notations are meaningful approximations of function that represent the time or space complexity of a program

① Big Oh (O) Notation

$f(n) = O(g(n))$

such that $|f(n)| \leq c\,|g(n)|$

if there exist a positive integer n_0 and a positive number c, for all $n \geq n_0$

Ex

$f(n)$	$g(n)$	
$16n^3 + 78n^2 + 2n$	n^3	$f(n) = O(n^3)$
$34n - 90$	n	$f(n) = O(n)$
56	1	$f(n) = O(1)$

$g(n)$ is the upper bound of the function $f(n)$

Omega (Ω) Notation

Def^n: $f(n) = \Omega(g(n))$

if there exists a positive integer n_0 and a positive number C such that

$|f(n)| \geq C\,|g(n)|$ for all $n \geq n_0$

Ex

$f(n)$	$g(n)$	
$16n^3 + 8n^2 + 2$	n^3	$f(n) = \Omega(n^3)$
$24n + 9$	n	$f(n) = \Omega(n)$

Theta (Θ) Notation

$f(n) = \Theta(g(n))$

two positive constant c_1 and c_2 and a positive integers n_0 such that

$c_1|g(n)| \leq |f(n)| \leq c_2|g(n)|$ for all $n \geq n_0$

Ex

$f(n)$	$g(n)$	
$28n$ ~~+6~~ 9	n	$f(n) = \Theta(n)$ Since $f(n) > 28n$ and $f(n) \leq 37n$ for $n \geq 1$

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$16n^2 + 30n - 90 \quad n^2 \quad f(n) = \theta(n^2)$

Solve

$f(n) = 16n^2 + 30n + 90$

we can drop it

$n_0 = 90$

$= 16n^2 + 30n + 9 \cdot n^2$

$c = 90$ 90 Zr

$= 16n^2 + 30 \cdot n \cdot n + 9 \cdot n^2$

$n^2 \geq 30n$

n_0

$16n^2 + 30n^2 + 9 \cdot n^2$

$55\, n^2$

$n_0 = 30$

$f(n) = O(g(n)) \quad c = 55$

$C. =$

$f(n) = 16n^2 + 30n + 90$

$= 16n^2$

$f(n) = O(g(n))$

$c = 16$

find the $g(n)$, n_0, c_1, c_2 for the g

$$f(n) = 28n + 9$$

$$g(n) = 28n + n$$

$$= 37n$$

$$37 \cdot g(n) \qquad n \geq 1$$

$$c \cdot g(n) \qquad c = 37$$

$$g(n) = 37\, g(n)$$

$$g(n) \leq c\, g(n)$$

$$g(n) = 28n + 9 \qquad n \geq 1$$

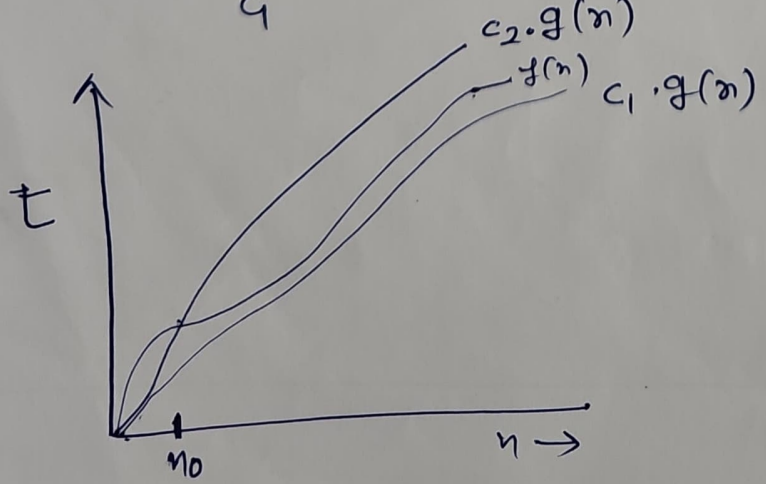
$$= 28n$$

$$c \cdot g(n)$$

$$g(n) \geq c\, g(n)$$

$$28\, g(n) \leq g(n) \leq 37 \cdot g(n) \qquad n \geq 1 = n_0$$

c_1 $\qquad$ c_2



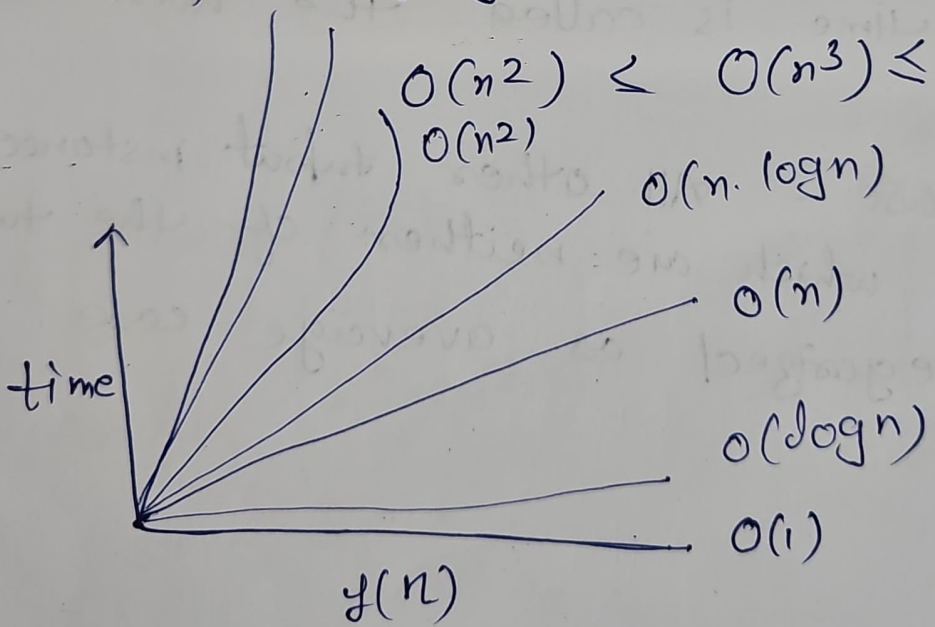
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Eg Solve $f(n) = 5n^3 + 16n^2 + 3n + 8$

$c_2 = 32$
$c_1 = 5$
$n_0 = 1$

Time Complexity

$O(1) < O(\log n) < O(n) < O(n \cdot \log n) \leq$
$O(n^2) \leq O(n^3) \leq O(2^n)$



Growth rat

Average, Best and Worst case Complexity

Linear Search

6 17 71 21 9 3 1 5 -23

Best Case - That input instance for which the algorithm takes the minimum possible time is called the best case.

Worst case - That input instance for which algorithm takes the maximum possible time is called the worst case

Average Case - All other input instances which are neither of the two are categorized as average case.

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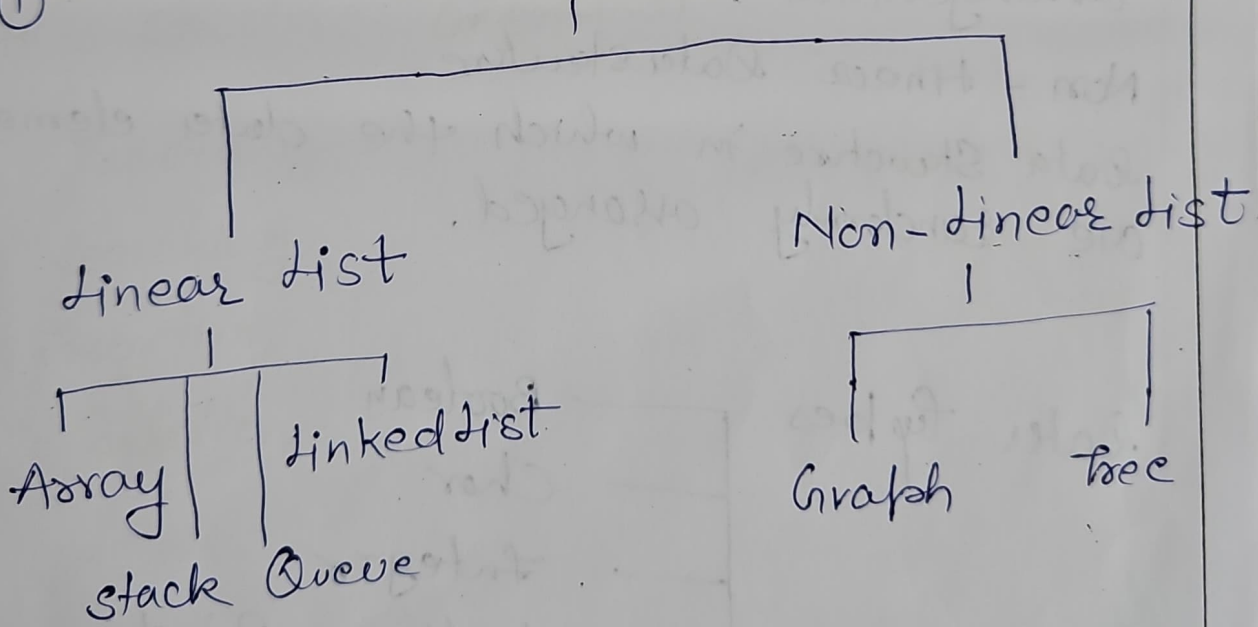
characteristics of Data Structure

Related Data + operation on data = Data Structure

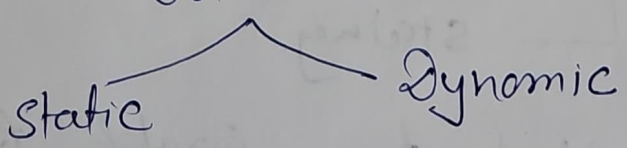
Data structure + Algorithm = Program

Type of Data Structure

① Data structure



② Data structure



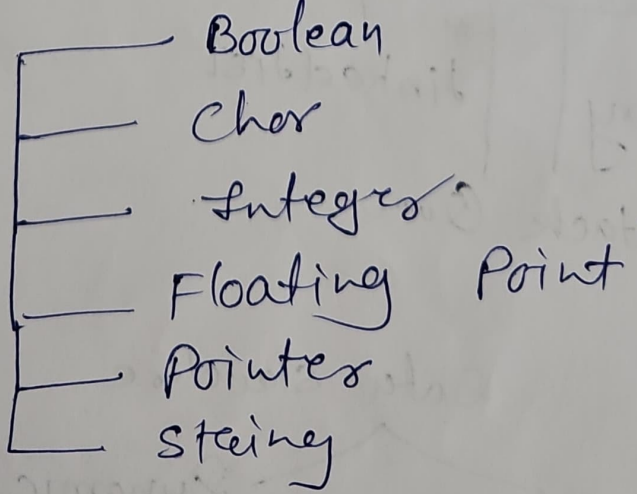
Static DS have fixed format and size along with memory locations.

Dynamic structures feature non-fixed memory capacities, shrinking or expanding as required by the program and its execution requirements.

Linear Data Structure -
Data element in a linear data structure are linked to one another in a sequential arrangement.

Non-linear Data structure -
Data structure in which the data elements are randomly arranged.

Data types



Data type — Abstract — Graph, Tree
— Composite — Array, class
— Primitive —

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Complexity - Space - time Trade-offs.

This refer to a choice blw algorithmic solution of a data processing problem that allow one to decrease the running time of an algorithmic solution by increasing the space to store data and vice - versa.

Data - structure Operation

1. Traversing
2. Searching
3. Insertion
4. Deletion

Array

A linear array is a list of a finite number n of homogeneous data elements

① element of array refer by index set

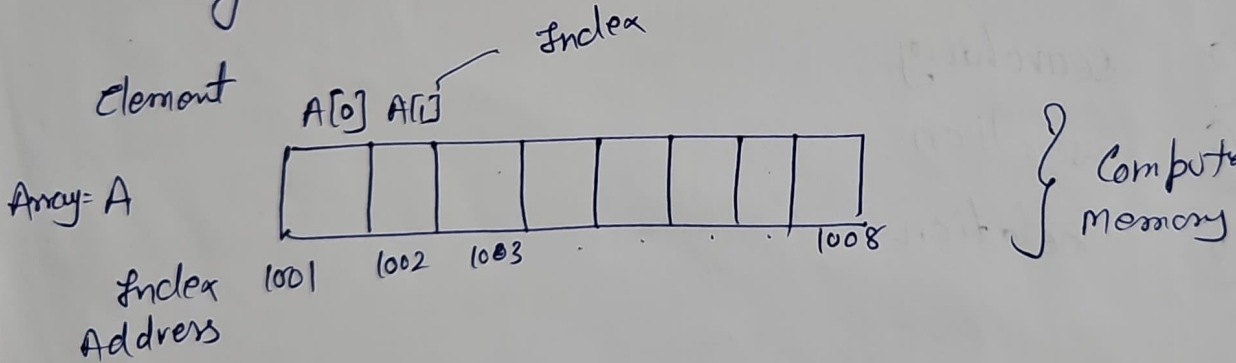
② element store in successive memory locations

$$\text{Length} = UB - LB + 1$$

UB - Upper Bound
LB - Lower bound

n is refer no. of element or length

Memory Representation of Array



int A[8]; // 'C' declaration

$$LOC(A[K]) = Base[A] + w(K - \text{lower bound})$$

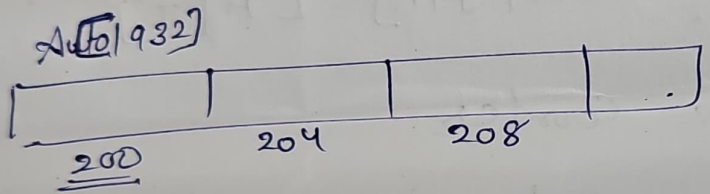
Base [A] — Address of first element

w — width

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Eg Base (A) = 200
w = 4 words per memory
K = 1965

A[1965] = Base + w[1965 − 1932]



= 200 + 4[33]

A[1965] = 332

Traversing linear Array

LA – Linear Array, LB – Lower Bound, UB –

1. K = LB
2. Repeat 3 and 4 while K ≤ UB
3. [Visit element] Apply Process to LA[K]
4. [Increase Counter] Set K = K+1
   End
5. Exit

Insertion and Deletion

Insert (LA, N, k, Item)

1. [Initialize Counter] Set J := N
2. Repeat step 3 and 4 while J ≥ k
3. [Move J element downward]
   LA[J+1] = LA[J]
4. Decrease Counter
   Set J = J-1

(End of loop)

5. Set LA[k] = Item
6. Set N = N+1
7. Exit

Delete (LA, N, k, Item)

1. set Item = LA[k]
2. Repeat J = k to N-1
   set LA[J] = LA[J+1]
3. Set N = N-1
4. Exit

Bubble Sort

Bubble (DATA, N)

1. Repeat step 2 and 3 for K=1 to N-1
2. Set PTR = 1
3. Repeat while PTR ≤ N-K
   (a) If Data[PTR] > Data[PTR+1] then
   Interchange Data[PTR] and Data[PTR+1]
   (b) Set PTR = PTR+1
   End of loop
4. Exit

Searching

① Linear Search (

Linear (DATA, N, Item, LOC)

1. Insert Item at end of Data
2. Set LOC = 1